A Design Application Handbook of Indirect Biophilic Attributes in Hawaii Hotel Interior Design

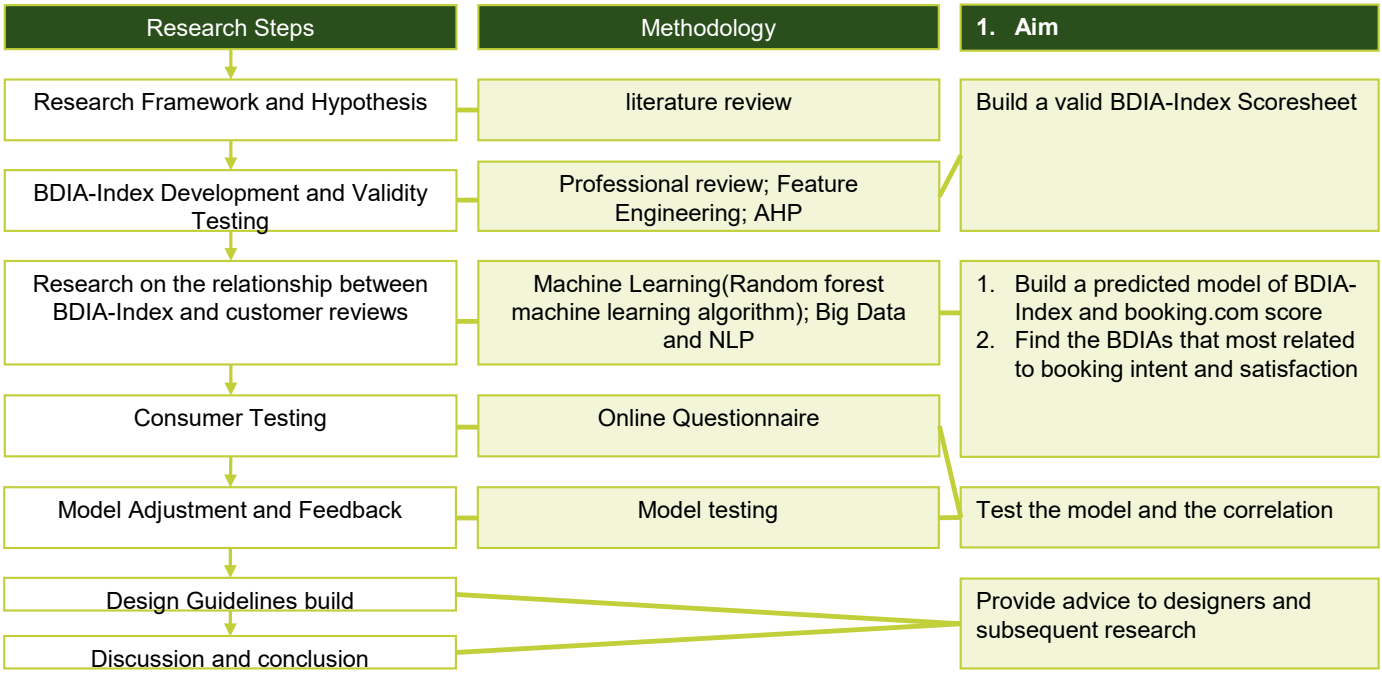
——Comparative Study on Waikiki 4-star hotels

Presenter: Chunya Wu
Organization: University of Hawaii at Manoa, ZGF Architects LLP
Contact: chunya.wu@zgf.com
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All texts, files and datas available on Researchgate

What is BDIA?

The Biophilic Design Indirect Attributes (BDIA) model is a quantitative assessment tool for evaluating the use of indirect biophilic attributes in hotel interiors, using a binary scoring system to quantify the indirect experience of nature. Developed through NLP analysis from more than 35,000 comments in Waikiki 4-star hotel, it integrates principles from the Design Quality Indicator (DQI) and the Biophilic Interior Design Matrix (BID-M) to offer a simple approach for predicting hotel evaluation scores on booking platform.

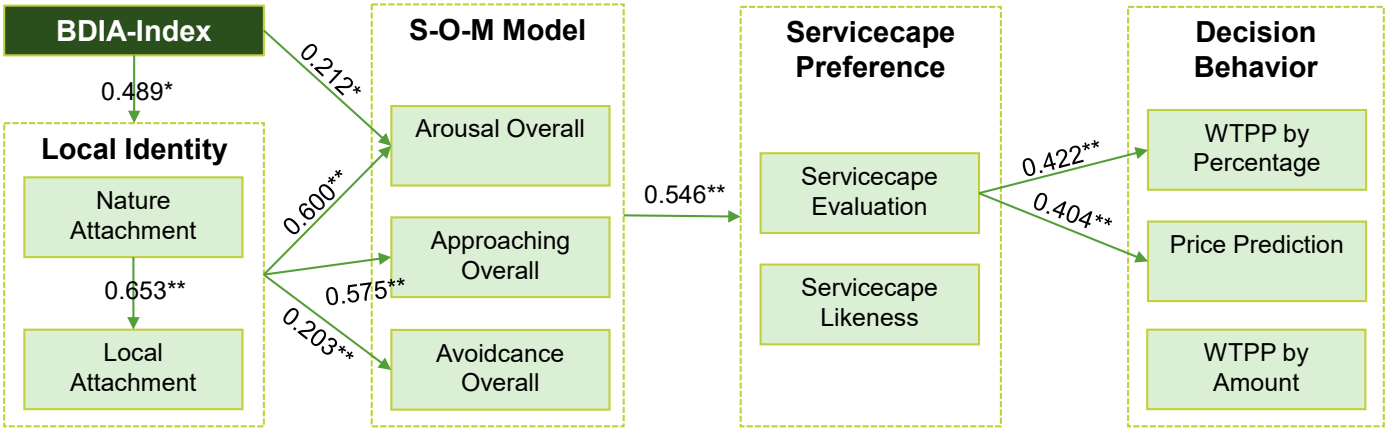
Fig. BDIA development process



How BDIA influence the customer's perception and behavior?

High BDIA directly increases customers' perception and sense of belonging to the "Hawaiian" degree of the design, as well as the degree to which they are aroused by the environment, and indirectly increases customers' willingness to approach the facilities and their love and evaluation of the facilities, thus affecting their willingness to pay.

Fig. BDIA's mechanism of action



Who can use this tool?

Architectural designers, interior designers, hotel managers, and owners can all predict the future performance of the project by referring to the design details.

What is included in BDIA? How can I apply?

BDIA contains 7 categories of measurement indicators from A to G. Each category is used to measure the proportion of an indirect biophilic design quality in the interior and consists of 3-6 subdivision indicators. These categories include Images of Nature, Natural materials, Natural colors, Simulating natural light and air, Naturalistic shapes and forms, Age/change/the patina of time, and Natural geometries. For a detailed description and usage introduction of each category and its subdivision indicators, please refer to the following article.

How to use BDIA in predicting my hotel's Booking.com ranking?

BDIA-Index is a tool that converts BDIA into a predicted Booking.com Overall Score. It contains 16 detailed questions about biophilic design, which you can find in the appendix. For each question, you need to evaluate your hotel in three scenarios (standard room, lobby, lounge) and answer yes (1) / not present (0). For definitions and detailed descriptions of elements. The scores from this table will be converted into scores using the following formula, with an explanatory power of 50.4%:

BDIA-Scenarios = SUM(Attribute*Weight)

For example,

BDIA-Guestroom=0.056*A01+0.03*A03+0.05*B01+0.06*B03+0.054*B04+0.103*B06+0.045*C01+0.052*C03+0.05*D01+0.061*D03+0.07*E03+0.03*F01+0.048*F02+0.18*G01+0.068*G02+0.042*G03

BDIA-Index = 0.368*Guestroom+0.285*Loundge+0.347*Lobby

Booking.com Overall Score Prediction =7.292 + 1.854*BDIA-Index

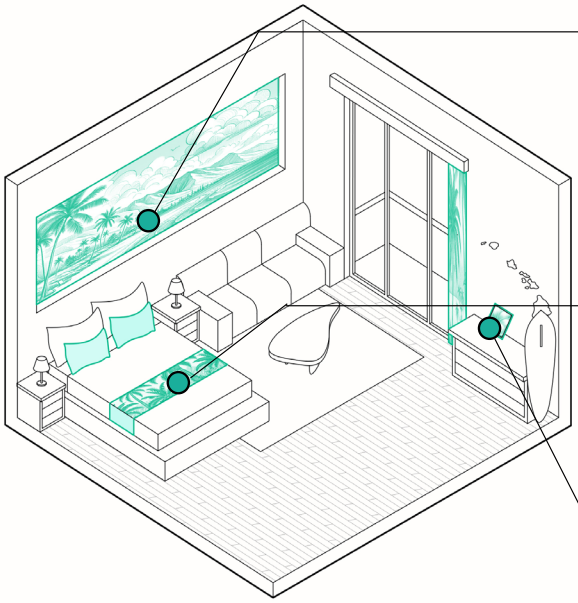
Glossary of terms

- **Recommand Level:** The recommended usage level is obtained by comprehensive analysis of customer questionnaires, machine learning, and big data analysis.
- **Frequency:** Proportion used in existing hotels.
- **↑/↓:** It will cause the corresponding indicator to rise(↑) or fall(↓).
- **Design Quality Perception:** Customers' perception of the quality of a certain design quality discovered through big data analysis. For example, "Form Design Quality Perception" refers to the quality perception of the design quality of "Form". The higher the perception, the more pleasant the hotel is perceived by customers, and is associated with more positive emotions and positive reviews.
- **Arousal Level:** The degree to which customers are psychologically aroused by certain factors. The higher the level of arousal, the more likely it is to produce positive environmental perceptions and experiences.
- **Local Identity Level:** The degree that customers think the hotel design style represents "Hawaii". The higher the perception, the more positive the customer's evaluation.
- **Servicecape Perception Level:** Customers' comprehensive perception of space satisfaction, luxury, and comfort.
- **WTPP Level:** The willingness of customers to pay an additional premium for this design.



A. Images of Nature

The image and representation of nature in the built environment—plants, animals, landscapes, water, geological features—can be both emotionally and intellectually satisfying. These images can occur through the use of photographs, paintings, sculptures, murals, video, computer simulations, and other representational means. Single or isolated images of nature typically exert little impact. Representational expressions of nature should be repeated, thematic, and abundant.



- Strong positive predictors
- Apotential design standard for high-end hotels
- Can continue to be used in future designs
- Can refer to existing design projects

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos).

Frequency
↑ Form Design Quality Perception
↑ Arousal Level
↑ Willing to Pay Premium



Carpets, pillows, curtains and other fabrics contain direct natural scenes (such as cartoon cat pillows)

Frequency
↑ Space Design Quality Perception



Use decorations with natural forms (such as ceramic rockeries).

Frequency
↑ Form Design Quality Perception

Examples



Aston Waikiki Beach Tower / 9.4
1. Using natural images in rooms is associated with positive perceptions of design quality
2. Natural images are the most commonly used BDIA by low-cost hotels



Queen Kapiolani Hotel/ 8.4
Multiple natural images have stronger effects than single locations

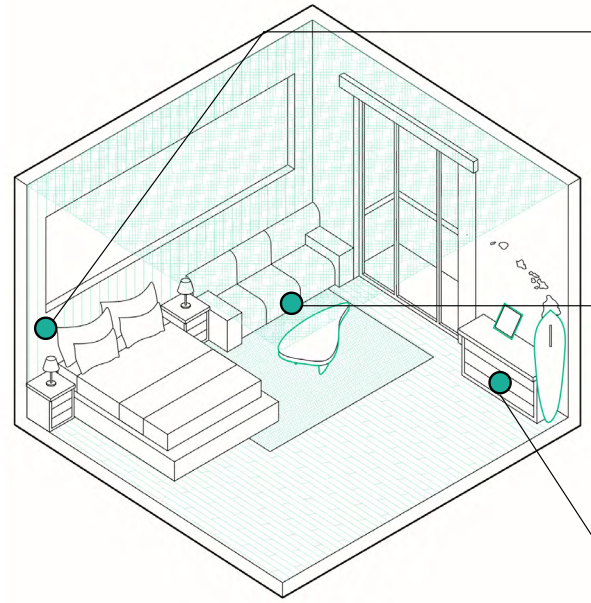


Sheraton Waikiki Beach Resort / 8.6
Paintings/artwork are associated with positive perceptions, prints/prints may be associated with negative perceptions



B. Natural materials

The transformation of materials from nature frequently elicits positive visual and tactile responses, which few artificial materials can duplicate. Prominent natural building and decorative materials include wood, stone, wool, cotton, and leather, used in a wide array of products, furnishings, fabrics, and other interior and exterior designs.



- may negatively correlated to quality perception
- associate with a lower level or outdated
- Widly used in all level hotels

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



Background walls, columns and other hard wall decorations are made of natural materials.

Frequency
↓ Decoration Design Quality Perception
↑ Servicecape Level



The ceiling is made of natural materials.

Frequency
↑ Space Design Quality Perception



Furnitures are made of natural materials.

Frequency
↑ Decoration Design Quality Perception

Examples



Hana-Maui Resort - Destination by Hyatt / 8.5
Natural material flooring (such as wooden flooring) will cause a decrease in quality perception and is associated with low cost and obsolescence. Carpet and slate floors will cause an increase in quality perception.



Halepuna Waikiki by Halekulani/ 9.0
Natural material ceilings are significantly associated with positive quality perceptions

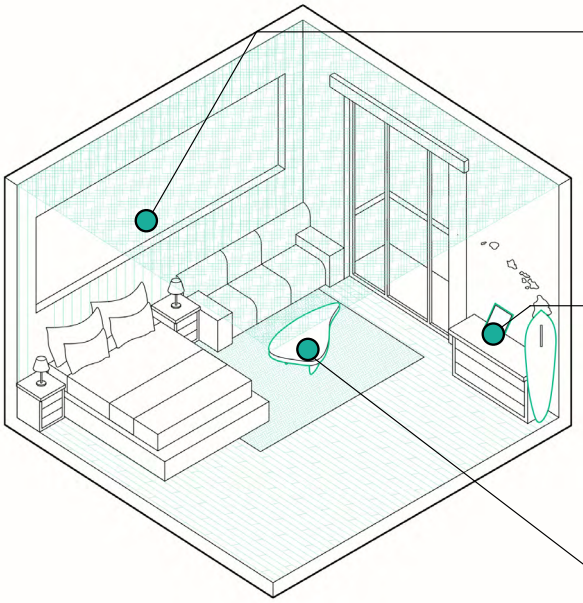


OUTRIGGER Waikiki Beach Resort / 8.7
Almost all Hawaii hotels (92.3%) use wooden furniture and decorations, which has become a design standard



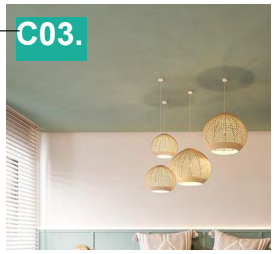
C. Natural colors

The effective biophilic application of color should generally favor muted “earth” tones characteristic of soil, rock, and plants. The use of bright colors should be cautiously applied, and emphasize such appealing environmental forms as flowers, sunsets and sunups, rainbows, and certain plants and animals. The occurrence of highly artificial, contrasting, and “vibrating” colors should be avoided.



- may negatively correlated to quality perception
- associate with a lower level or outdated
- Widly used in all level hotels

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



The ceiling is made of natural colors.

Frequency
↑ Arousal Level
↑ Local Identity Level



Decorations are made of natural colors.

Frequency
↑ Arousal Level



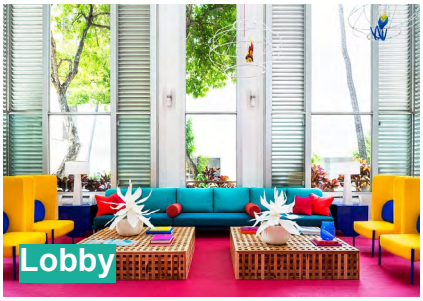
Furnitures are made of natural colors.

Frequency
↑ Arousal Level
↑Servicecape Perception Level

Examples



Prince Waikiki / 8.8
1. Rooms rated best in soft beige or Morandi colors
2. Intense colors are associated with lower price perceptions and higher discomfort perceptions



Shoreline Hotel Waikiki / 8.1
1. The use of strong colors in the lobby is associated with positive “Vibe” and “Atmosphere” ratings
2. The use of strong colors in public spaces increases perceived spatial quality and WTPP

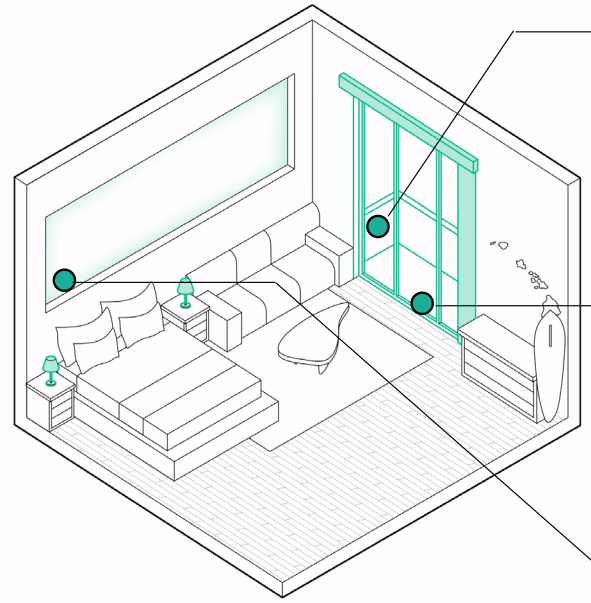


Marriott's Kaua'i Beach Club / 8.4
Using local color matching can increase the perception of "Local identity", thereby enhancing quality perception and booking intention



D. Simulating natural light and air

Indoor lighting and processed air have been made possible by advances in building technology and construction. The trade-of has often been the occurrence of static conditions that can be physically and psychologically debilitating. Artifcial light can be designed to mimic the spectral and dynamic qualities of natural light. Processed air can also simulate qualities of natural ventilation through variations in airfow, temperature, humidity and barometric pressure.





Adjustable, openable sash windows allow for natural ventilation.

Frequency

↑ Form Design Quality Perception



Adjustable interior or exterior sun visors or blinds, or balconies.

Frequency



Use indirect ambient lighting.

Frequency

- may not directly correlated to quality perception
- Widly used in high-end hotels
- Balcony is postively related to comments

Recommand Level

4-Star Hotels Frequency

Other Hotels Frequency

Examples



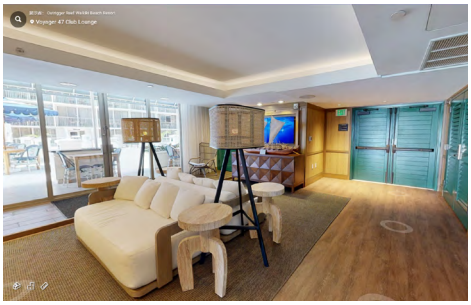
Sheraton Waikiki Beach Resort / 8.6

1. Balcony is a postive element, but it may also be related to thermal discomfort.
2. The direction of the light source is irrelevant to the evaluation.

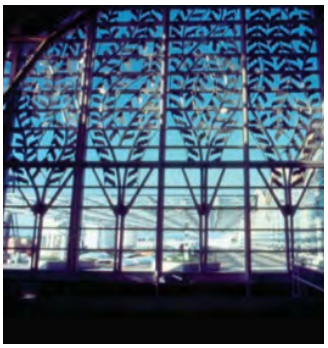


Halepuna Waikiki by Halekulani/ 9.0

Whether public spaces can be ventilated is irrelevant to the evaluation.

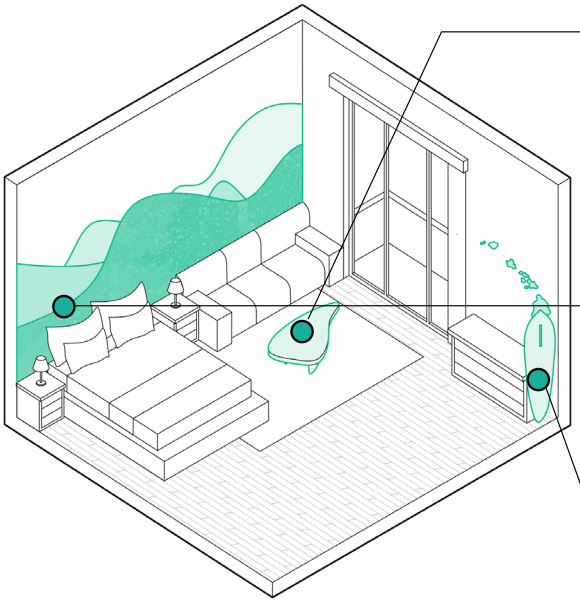


OUTRIGGER Reef Waikiki Beach Resort / 8.7



E. Naturalistic shapes and forms

The experience of shapes and forms characteristic of the natural world can be especially appealing. These naturalistic forms can be extraordinarily diverse from the leaf-like patterns found on columns, the shapes of plants on building facades, to animal facsimiles woven into fabrics and coverings. The occurrence of naturalistic shapes and forms can transform a static space into one that possesses the dynamic and ambient qualities of a living system.



- Indirectly correlated to quality perception
- Rarely used in hotel design

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



The furniture adopts irregular natural forms.

Frequency

↓ Form Design Quality Perception
↑ Style Design Quality Perception
↑ Arousal Level
↑ Serviceape Perception



Hard decoration (walls, ceilings, etc.) adopts irregular natural forms.

Frequency

↑ Arousal Level
↑ Willing to Pay Premium



The decorations adopt irregular natural forms.

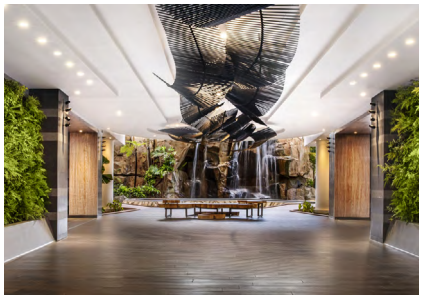
Frequency

↑ Arousal Level
↑ Local Identity Level
↑ Serviceape Perception

Examples



The Westin Maui Resort & Spa, Ka'anapali / 9.4
Naturalistic shapes and forms in guestroom is either associated with negative outcome scores or not associated with results in all analyses, and should be avoided in design as much as possible.



The Westin Maui Resort & Spa, Ka'anapali / 9.2

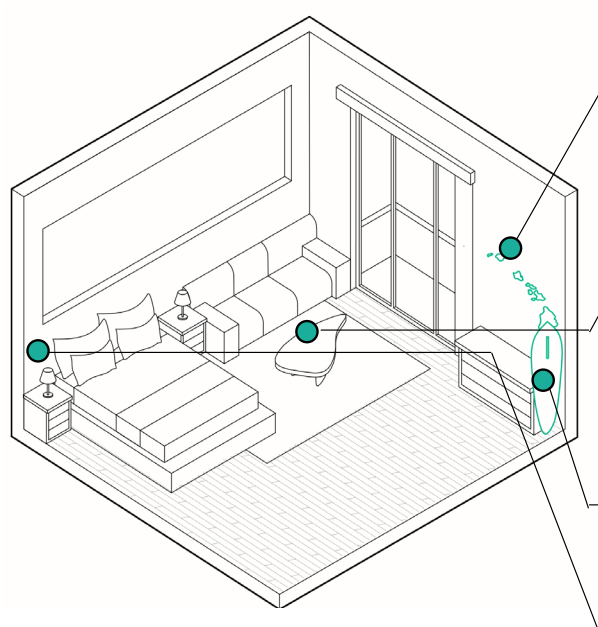


Hyatt Centric Waikiki Beach / 8.6



F. Age, change, and the patina of time

People respond positively to these dynamic forces and the associated patina of time, revealing nature's capacity to respond adaptively to ever-changing conditions. These dynamic tendencies are often most satisfying when balanced by the complementary qualities of unity and stability. Change and a patina of time can be achieved through such design strategies as naturally aging materials, weathering, a sense of the passage of time, and in other ways.



- significantly correlation with positive rankings
- a new design potential opportunity
- can be more actively adopted in future designs

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



There are nostalgic elements in the decoration style (such as 90s posters).
Frequency
↑ Atmosphere Design Quality Perception



The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).
Frequency
↑ Design/Atmosphere/Decoration Design Quality Perception
↑ Arousal Level
↑ Local Identity Level
↑ Serviceape Perception



Use recycled materials, furniture or decor.
Frequency
↑ Design/Atmosphere/Form Design Quality Perception
↑ Local Identity Level



Use materials that will show natural weathering or aging (such as copper).
Frequency
↑ Design/Atmosphere/Decoration/Style Design Quality Perception
↑ Local Identity Level

Examples



Wayfinder Waikiki / 8.5
If the design is considered local, retro or nostalgic style, it will increase its positive evaluation. But if the customer think it's outdated or worn, their perception will drop.



Wayfinder Waikiki / 8.5
Local Identity is related to higher Booking scores, higher Quantitative Feeling And Evaluation, and more positive consumption decisions.

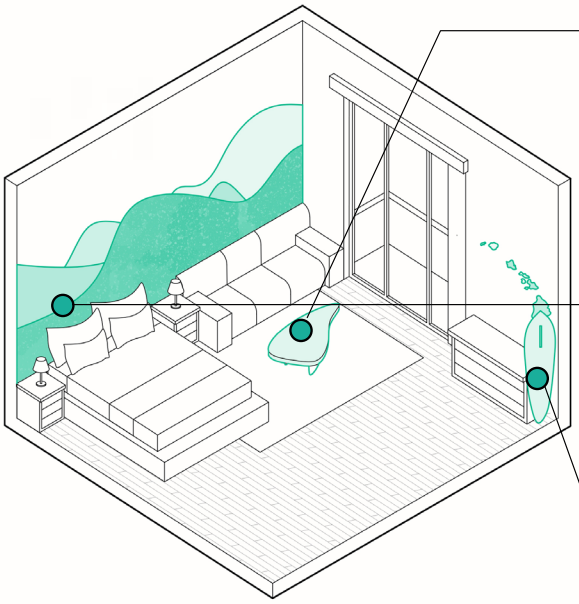


Wayfinder Waikiki / 8.5
Age, change, and the patina of time are the strongest predictors of Postive Comments Emotional Tendency. It may improve overall perception and evaluation by enhancing customers' local identity.



G.Natural geometries

The satisfying experience of nature can also be revealed through imaginative and fantastic depictions. For example, the “wings” of the Sydney Opera House suggest the qualities of a bird; Notre Dame’s stained glass windows, a rose-like fower; while, the skyline of some cities mimic the vertical heterogeneity of a forest. None of these designs actually occurs in nature, but they all draw from design principles and characteristics of the natural world.



- Strong positive predictors
- Apotential design standard for high-end hotels
- Can continue to be used in future designs
- Can refer to existing design projects

Recommand Level
4-Star Hotels Frequency
Other Hotels Frequency



The furniture adopts abstract and geometric patterns that imitate nature.

Frequency
↑ Decoration Design Quality Perception
↑ Arousal Level



Hard decoration (walls, ceilings, etc.) adopts abstract and geometric patterns that imitate nature.

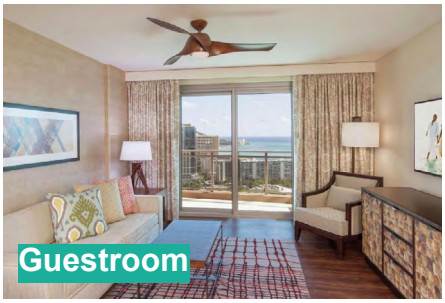
Frequency
↑Form Design Quality Perception
↑ Arousal Level



Decorations and soft furnishings adopt abstract and geometric patterns that imitate nature.

Frequency
↑Form Design Quality Perception
↑ Serviceape Perception

Examples



Hilton Grand Vacations Club Grand Waikikian Honolulu / 8.8
Using room decorations with natural textures can significantly improve perceived design quality and environmental satisfaction



Aston Waikiki Beach Tower / 9.4
Can be harmoniously combined with BDIA: natural images



The Surfjack Hotel & Swim Club / 8.7

Appendix: BDIA Scoresheet

You can access the automatic calculation form by following the link. Please copy it into your Excel for use.

https://docs.google.com/spreadsheets/d/137NjsYtomJauL6hnKsDD50ldK0LIHkyMszzRJh8lZE8/edit?usp=sharing

BDIA		Weight	Guestroom	Lobby	Lounge
Images of nature					
A01	Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos).	0.056			
A03	Use decorations with natural forms (such as ceramic rockeries).	0.03			
Natural materials					
B01	Background walls, columns and other hard wall decorations are made of natural materials.	0.05			
B03	The ceiling is made of natural materials.	0.06			
B04	The floors are made of natural materials.	0.054			
B06	Decorations are made of natural materials.	0.103			
Natural colors					
C01	The hard decoration of the background wall, pillars and other walls are in natural colors.	0.045			
C03	The ceilings are in natural colours.	0.052			
Simulating natural light and air					
D01	Adjustable sash windows allow for natural ventilation.	0.05			
D03	Use indirect ambient light.	0.061			
Naturalistic shapes and forms					
E03	The decorations adopt irregular natural forms.	0.07			
Age, change, and the patina of time					
F01	There are nostalgic elements in the decoration style (such as 90s posters).	0.03			
F02	The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).	0.048			
Natural geometries					
G01	The furniture adopts abstract and geometric patterns that imitate nature.	0.18			
G02	Hard decoration (walls, ceilings, etc.) adopts abstract and geometric patterns that imitate nature.	0.068			
G03	Decorations and soft furnishings adopt abstract and geometric patterns that imitate nature.	0.042			
BDIA-Scenarios					
BDIA-Index					
Booking.com Overall Score Prediction					

BDIA-Scenarios = SUM(Attribute*Weight)

Example: BDIA-Guestroom=0.056*A01+0.03*A03+0.05*B01+0.06*B03+0.054*B04+0.103*B06+0.045*C01+0.052*C03+0.05*D01+0.061*D03+0.07*E03+0.03*F01+0.048*F02+0.18*G01+0.068*G02+0.042*G03

BDIA-Index = 0.368*Guestroom+0.285*Loundge+0.347*Lobby

Booking.com Overall Score Prediction =7.292 + 1.854*BDIA-Index

Appendix: BDIA Checklist

This checklist helps you quickly check how each BDIA relates to perceived design quality. For example, you can examine which BDIA contributes to the perceived quality of a specific space (e.g. lobby, guest room).

Attributes	Correlation	Guestroom	Lobby	Lounge	Design	Atmosphere	Decoration	Form	Furniture	Space	Style	S-O-M Model	Local Identity	Servicecape	WTPP
F02	●	◎	◎	◎	●	●	●	◎	◎	◎	◎	●	●	●	◎
F04	●	◎	●	◎	●	●	●	◎	◎	◎	●	◎	●	◎	◎
G03	●	◎	◎	◎	◎	◎	●	◎	●	◎	◎	◎	◎	●	◎
A01	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	●	◎	◎	●
A03	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
B06	●	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
E03	●	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	●	●	◎
F01	●	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎	◎
F03	●	◎	◎	◎	●	●	◎	●	◎	◎	◎	◎	●	◎	◎
G01	●	◎	◎	○	◎	◎	●	◎	◎	◎	◎	●	◎	◎	◎
B03	●	◎	◎	◎	◎	○	◎	◎	◎	●	◎	◎	◎	◎	◎
C03	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	●	◎	◎
C01	●	◎	○	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎	◎
D01	●	◎	◎	○	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
G02	●	○	◎	◎	◎	◎	◎	●	○	◎	◎	●	◎	◎	◎
A02	●	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
C05	●	◎	●	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎
C02	●	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
D03	●	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎	◎
B02	◎	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎
B05	●	◎	◎	◎	◎	◎	●	◎	○	◎	◎	◎	◎	◎	◎
B01	●	◎	○	◎	◎	◎	○	◎	◎	◎	◎	◎	◎	●	◎
C06	●	●	○	◎	◎	◎	◎	◎	◎	◎	◎	●	◎	●	◎
D02	◎	◎	◎	◎	◎	◎	◎	●	◎	◎	◎	◎	◎	◎	◎
E01	●	◎	◎	○	◎	◎	◎	○	◎	◎	●	●	◎	●	◎
D04	●	◎	◎	◎	◎	◎	◎	◎	○	◎	◎	◎	◎	◎	◎
C04	●	●	◎	◎	◎	◎	○	◎	◎	◎	◎	◎	◎	◎	◎
B04	●	◎	○	○	○	○	○	◎	◎	◎	○	◎	◎	◎	◎
E02	◎	◎	◎	◎	◎	◎	◎	◎	○	◎	◎	●	◎	◎	●

- **Scenarios (Geustroom | Lobby | Lounge):** The correlation between customer positive evaluation and specific scenarios.
- **Quality Perception (Design | Atmosphere | Decoration | Form | Furniture | Space | Style):** Customers' perception of the quality of a certain design quality discovered through big data analysis. For example, "Form" refers to the quality perception of the design quality of Form.
- **S-O-M Model:** The degree to which customers are psychologically aroused by certain factors. The higher the level of arousal, the more likely it is to produce positive environmental perceptions and experiences.
- **Local Identity:** The degree that customers think the hotel design style represents "Hawaii".
- **Servicecape:** Customers' comprehensive perception of space satisfaction, luxury, and comfort.
- **WTPP:** The willingness of customers to pay an additional premium for this design.

○ Negatively correlated with the analysis results
◎ No correlation with the analysis results
● Positively correlated with the analysis results